

***THREEFOLD SOURCES SOUGHT FOR
PRODUCT(S) MADE IN AMERICA & SMALL BUSINESS MANUFACTURER(S)
THIS NOTICE IS FOR MARKET RESEARCH PURPOSES ONLY
NOT A REQUEST FOR PROPOSAL OR SOLICITATION**

Sources Sought Number: 75F40126SS132810

Information regarding any technical point of contact will not be provided and no appointments for presentations will be made.

Notice of Intent

Responses to this Sources Sought announcement will assist the Government in determining whether any future requirements similar to this one should be set aside for small business, made available to full and open competition or procure through sole-source acquisition procedures. Interested Parties shall respond with capability statements which are due by email to howard.nesmith@fda.hhs.gov by **1:00 PM (Central Time), April 6, 2026, Referencing 75F40126SS132810.**

The U.S. Food and Drug Administration (FDA), Human Foods Program (HFP), Office of Laboratory Operations and Applied Science (OLOAS) consisting of the Office of Applied Microbiology and Technology (OAMT) and Office of Chemistry and Toxicology (OCT), requires the acquisition of multiple Food Safety Real-Time Polymerase Chain Reaction (PCR) System.

***Part One**

Product(s) Made In America

Executive Order 14005 of January 25, 2021, was issued Ensuring the Future is Made in All of America by All of America's Workers, which directs Federal agencies to maximize the use of goods, products, and materials produced in, and services offered in, the United States through their financial assistance awards and procurements.

In compliance with Executive Order 14005 the FDA is seeking potential provider(s) who can provide and install an Electro-Mechanical Dynamic Load Frame System which is **Made in America**. Regardless of North American Industry Classification System (NAICS) Code Business Size Standards all interested parties who can provide the product(s) described herein **Made in America** may respond to this Notice.

***Part Two**

Small Business Manufacturer(s)

Potential provider(s) considered a **Small Business** under North American Industry Classification System (NAICS) 334516 – Analytical Laboratory Instrument Manufacturing; with a Small Business Size Standard of 1,000 employees and is capable of providing the FDA with the product(s) described herein and **Made in America** may respond to this Notice.

***Part Three**

Product(s) Made Outside America

Potential provider(s) no matter their North American Industry Classification System (NAICS) size who can provide the product(s) described herein which is made **Outside America** may also respond to this Notice.

All responses shall be submitted via email to Howard.Nesmith@fda.hhs.gov and provide:

- Proposed Product Information and Description
- Capability Statement which indicates proposed product(s) is Made in America
- Business Name
- Unique Entity ID (SAM) Number
- Business Contact Information to include point of contact name, mailing address, phone number and email address
- NAICS Small Business Program Representations (Small or Large)
- **Country of Origin of proposed product(s) if Made Outside America**

Background:

The FDA/OAMT & OCT both optimize and validate rapid and highly sensitive microbiological and molecular methods and instrumentation used for recovery, detection, and quantification of enteric bacterial pathogens and their related toxins from foods and cosmetics. The ORTS laboratory network performs scientific testing on products which are regulated by FDA as well as related food processing and farm production environments. To ensure that the testing results are competent and suitable for regulatory purposes, the ORTS laboratories utilize known validated methods including Real-Time PCR protocols according to the reference methods outlined in the FDA Bacteriological Analytical Manual (BAM) for the detection of pathogens, including Salmonella, Shiga-toxin Escherichia coli, Listeria, Hepatitis A/Norovirus and Cyclospora analysis.

Minimum Technical Requirements:

Any catalog number, manufacturer brand name, or proprietary description used in this solicitation is used as descriptive only, not restrictive, and used to indicate type, function, or desired quality. System shall be newly manufactured, latest model, not used, refurbished, or previously used for demonstration.

System Compatibility

- System shall be fully compatible with current validated Applied Biosystems 7500 Fast Real-Time PCR System assays without requiring assay revalidation or modification
- System shall be capable of detecting a minimum of six (6) distinct targets simultaneously within a single reaction

Optical System

- Minimum of 6 decoupled excitation and emission filter channels (96-well format configuration) multi-channel detection
- Filter combinations capability to support at least 21 distinct filter combinations to enable comprehensive multiplexing applications
- Factory calibration for the following fluorescent dyes at minimum: FAM, SYBR Green, VIC, ABY, NED, TAMRA, JUN, ROX, Mustang Purple, and Cy5 Dye Compatibility
- Capable of detecting target quantity differences as low as 1.5-fold in singleplex reactions sensitivity
- Minimum dynamic range of 10 logs (10 orders of magnitude) dynamic range

Thermal Block

- System shall include a sample format of 96-well, 0.1ml reaction block configuration to accommodate standard laboratory throughput requirements
- Sample format options compatible with both 96-well (0.1 mL and 0.2 mL reaction volumes) and 384-well block formats
- 96-Well system shall provide the capability to partition the block into six (6) independent temperature control zones, enabling simultaneous execution of multiple experiments with different thermal protocols
- Shall have an operational temperature range of 4°C to 99.9°C
- Heating/cooling method shall be a peltier-based thermal cycling system
- Maximum temperature variation of $\pm 0.4^{\circ}\text{C}$ across the block temperature uniformity
- Shall have ramp rate performance of:
 - Maximum block ramp rate: $\geq 6.5^{\circ}\text{C}/\text{second}$ (96-well, 0.2 mL format)
 - Maximum block ramp rate: $\geq 9.0^{\circ}\text{C}/\text{second}$ (96-well, 0.1 mL format)
 - Average sample ramp rate: $\geq 3.66^{\circ}\text{C}/\text{second}$

Connectivity and Data Management

- Shall support USB, Wi-Fi, and LAN network connections
- Shall be capable of connecting to manufacturer's cloud platform for remote monitoring, data analysis, and secure data sharing
- Shall have encrypted data transmission and storage capabilities for data security

User Interface Requirements

- Shall have at minimum 8.5-inch interactive touchscreen interface display
- Shall have pre-optimized, simplified protocol templates for standard applications protocol templates
- Shall have interface design suitable for operation by personnel with varying levels of user experience

Instrument Software Regulatory Compliance Features

- Shall have multi-level user access management controls and authentication
- Shall have comprehensive electronic audit trail functionality
- Shall have E-signature capability compliant with 21 CFR Part 11 requirements
- Shall have data integrity features which supporting FDA data integrity guidelines (ALCOA+ principles)

Performance Requirements

- Throughput run time capable of completing thermal cycling in less than 30 minutes for 96-well 0.2 mL block format using fast chemistry protocols
- Throughput sample capacity configurable for 96-well or 384-well formats as specified
- System shall be designed for continuous uptime laboratory operation with minimal downtime
- Dye calibration interval requirements shall not exceed once every two years under normal operating conditions
- Shall be designed for minimal routine maintenance requirements

System Computer Requirements

If procured in PC-attached configuration, the accompanying computer system shall meet or exceed the following:

- Intel Core i5 or equivalent/superior Processor
- 16 GB RAM minimum Memory
- Windows 10 (or current FDA-approved Windows OS version) Operating System
- Adequate storage capacity for data archiving per FDA requirements

Acceptance IQ/OQ/PQ Criteria

- **Installation Qualification (IQ)**
 - Proper installation per manufacturer specifications
 - Confirmation that installation site is suitable per above Instrument Software Regulatory Compliance Features
 - All specified hardware components present and functional
 - Network connectivity operational
 - Software installation complete with all regulatory compliance features enabled
 - Electrical safety verification and grounding confirmation
 - Shall provide documentation confirming (IQ)
- **Operational Qualification (OQ)**
 - Temperature uniformity within $\pm 0.4^{\circ}\text{C}$ specification
 - Ramp rate performance meeting specified values
 - Optical detection across all specified dye channels
 - Cloud connectivity and data transfer functionality
 - 21 CFR Part 11 compliance features operational
 - Shall demonstrate (OQ)
- **Performance Qualification (PQ)**
 - The vendor shall provide validation data demonstrating:
 - Sensitivity meeting 1.5-fold detection specification
 - Dynamic range of 10 logs
 - Run time performance for fast cycling protocols
 - Reproducibility across independent temperature zones
 - Shall provide validation data demonstrating (PQ)

Complete System Shall Include:

- Shall be delivered with all necessary supplies and accessories required for installation and start-up
- One (1) laptop computer if procured in PC-attached configuration meeting above System Computer Requirements
- User manuals and technical documentation
- All necessary consumables for IQ/OQ/PQ testing
- IQ/OQ/PQ protocols and executed documentation
- Training for FDA laboratory personnel (up to 4 users)
- At minimum one-year manufacturer warranty from date of installation

FOB Destination & Place of Performance:

U.S. Food and Drug Administration
Human Foods Program (HFP)
Multiple Laboratory Locations Across the United States

Respondents shall furnish sufficient technical information necessary for the Government to conclusively determine that the offered product/services meet the General Requirements/Specifications identified herein. Though the target audience is Small Business Manufacturers or Small Businesses capable of supplying services of a Small Business, all interested parties may respond. At a minimum, responses shall include the following:

- Business name, DUNS number, business address, business website, business size status (i.e., SB, VOSB, SDVOSB, HUBZone SB, SDB, WOSB, LB), point of contact name, mailing address (if different from business address), phone number and email address (Provide this same information if responding to provide a product manufactured by another firm);
- Sufficient descriptive literature that unequivocally demonstrates that offered service can meet the herein General Requirements/Specifications. All descriptive material necessary for the government to determine whether the offered meets General Requirements/Specifications including technical specifications, descriptive material, literature, brochures and other information, which demonstrates the capabilities of the respondent to meet the General Requirements/Specifications;
- If applicable, identification of the respondent's GSA Schedule contract(s) by Schedule number and SINs that are applicable to this potential requirement. Detailed list of items included and not included on the GSA contract to meet this General Requirements/Specifications;
- If a large business, identify the subcontracting opportunities that would exist for small business concerns;
- Though this is not a request for quote, informational pricing is encouraged.

The government is not responsible for locating or securing any information, not identified in a response.

The Government encourages any comments and/or suggestions from any interested party, regarding the specifications. While the Government will not respond directly to your comments and/or suggestions, we will consider them as we finalize the specifications in preparation for the forthcoming solicitation.

Disclaimer and Important Notes

This notice does not obligate the Government to award a contract or otherwise pay for the information provided in response. The Government reserves the right to use information provided by respondents for any purpose deemed necessary and legally appropriate. Any organization responding to this notice shall ensure that its response is complete and sufficiently detailed to allow the Government to determine the organization's qualifications to perform the work. Respondents are advised that the Government is under no obligation to acknowledge receipt of the information received or provide feedback to respondents with respect to any information submitted. After a review of the responses received, a pre-solicitation synopsis and/or a solicitation may be published in SAM.gov. However, responses to this notice will not be considered adequate responses to a solicitation.

Confidentiality

No proprietary, classified, confidential, or sensitive information shall be included in your response.

The Government reserves the right to use any non-proprietary technical information in any resultant solicitation(s).